

Comprehensive Model Results Summary

Hedonic Price Index Analysis

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14 Models: 7 Types $\times$ 2 Time Dimensions (Quarterly/Annual)

Quarterly Models - Cumulative Jevons Indices

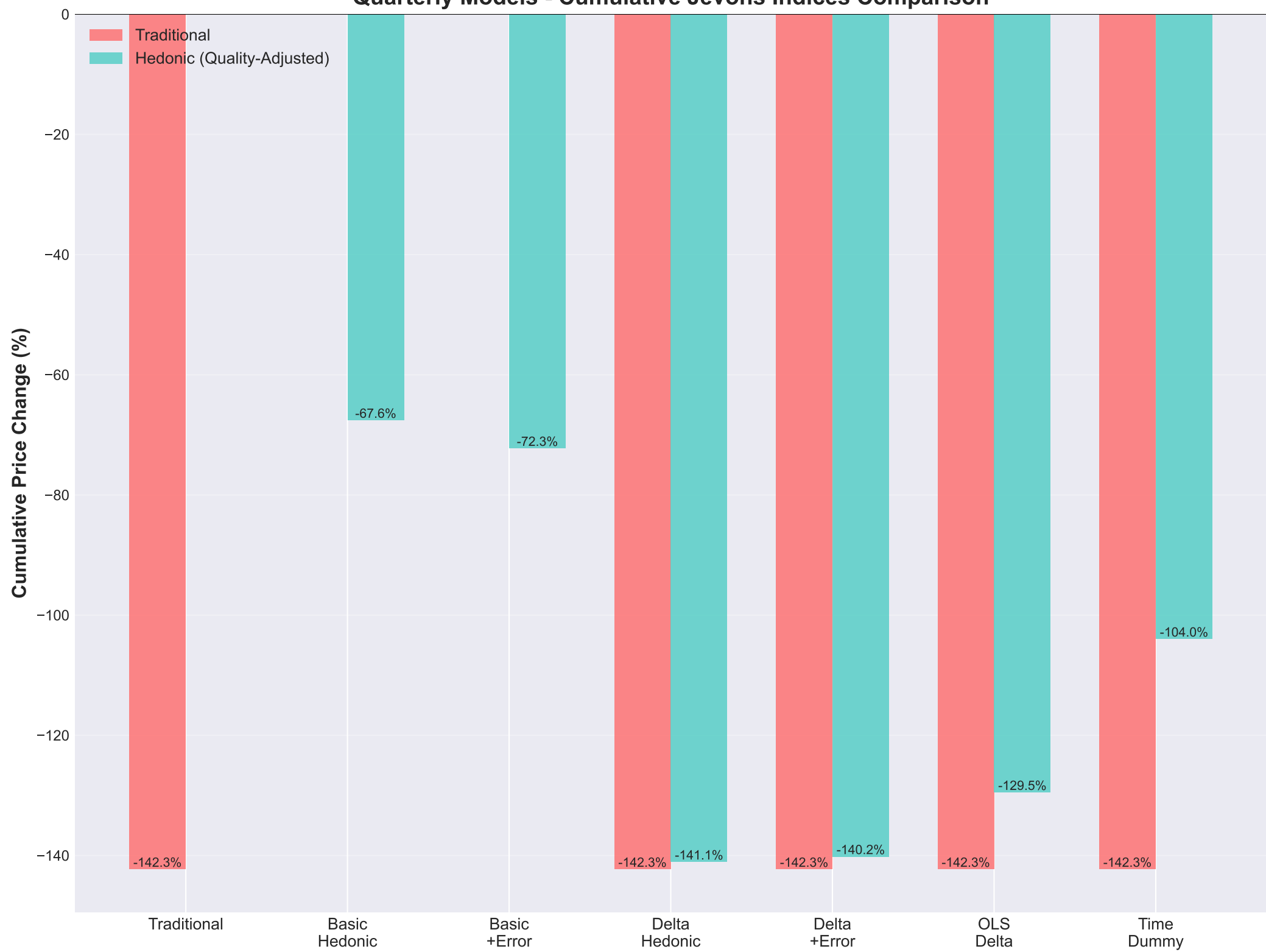
Model	Traditional	Hedonic	R ² (avg)	Description
Traditional Jevons Index	-1.4231 (-142.31%)	N/A	N/A	No quality adjustment (2020 Q1+)
Basic Hedonic	N/A	-0.6759 (-67.59%)	0.5745	Standard hedonic regression
Basic + Error Feature	N/A	-0.7226 (-72.26%)	0.7689	With error feature
Delta Model	-1.4231 (-142.31%)	-1.4110 (-141.10%)	0.2559	Direct price change modeling
Delta + Error	-1.4231 (-142.31%)	-1.4020 (-140.20%)	0.2772	Delta with error feature
OLS Delta Model	-1.4231 (-142.31%)	-1.2952 (-129.52%)	0.4364	OLS regression (no regularization)
Time Dummy Model	-1.4231 (-142.31%)	-1.0399 (-103.99%)	0.6597	Pooled data + time dummy

Annual Models - Cumulative Jevons Indices

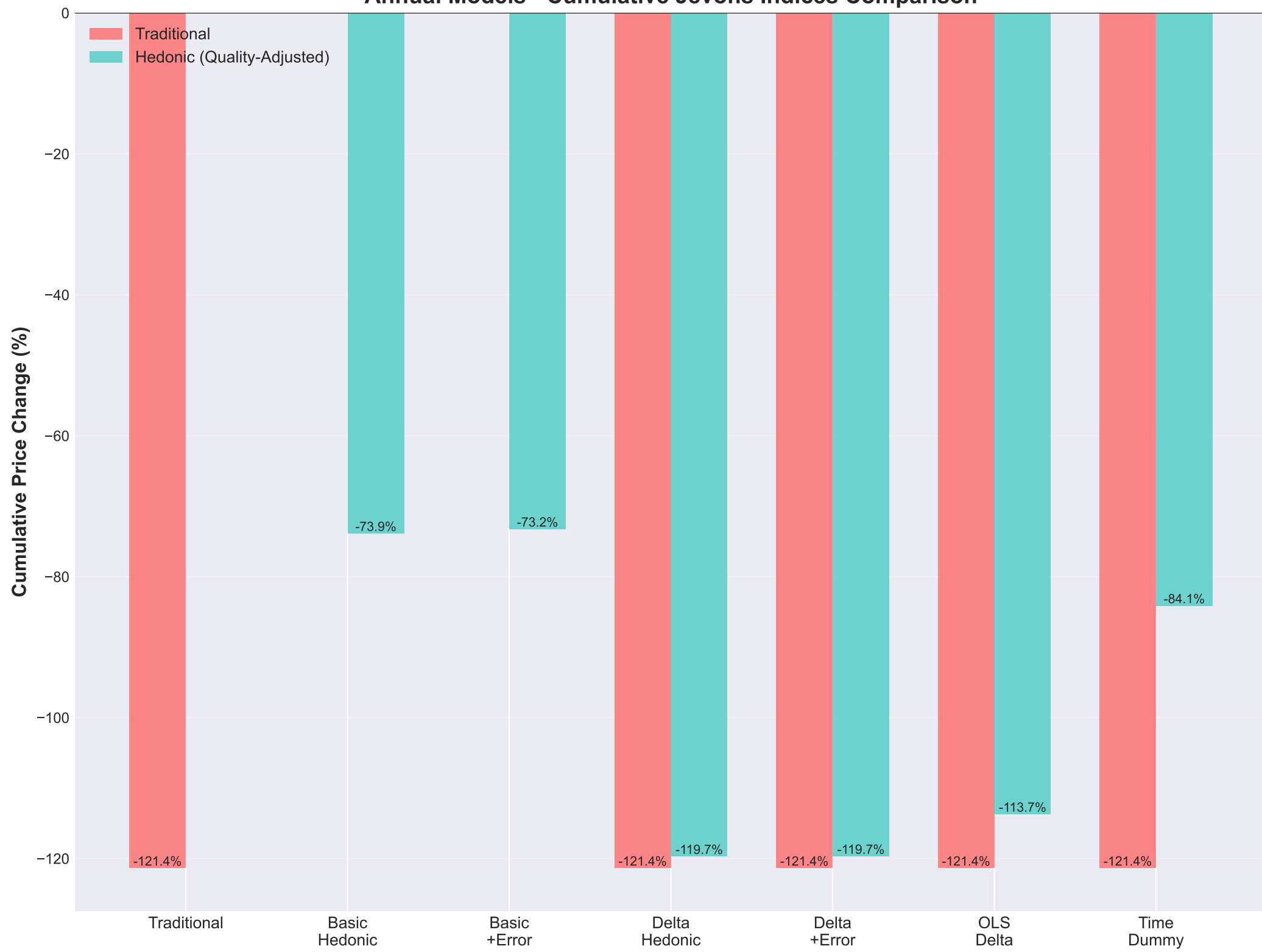
All models cover 2020 to 2025 (5 periods)

Model	Traditional	Hedonic	R ² (avg)	Description
Traditional Jevons Index	-1.2135 (-121.35%)	N/A	N/A	No quality adjustment (2020+)
Basic Hedonic	N/A	-0.7387 (-73.87%)	0.6330	Standard hedonic regression
Basic + Error Feature	N/A	-0.7325 (-73.25%)	0.7564	With error feature
Delta Model	-1.2135 (-121.35%)	-1.1970 (-119.70%)	0.2498	Direct price change modeling
Delta + Error	-1.2135 (-121.35%)	-1.1968 (-119.68%)	0.2623	Delta with error feature
OLS Delta Model	-1.2135 (-121.35%)	-1.1371 (-113.71%)	0.4226	OLS regression (no regularization)
Time Dummy Model	-1.2135 (-121.35%)	-0.8412 (-84.12%)	0.6225	Pooled data + time dummy

Quarterly Models - Cumulative Jevons Indices Comparison



Annual Models - Cumulative Jevons Indices Comparison



Model Descriptions

- **Traditional Jevons Index:**

Direct calculation using actual prices. No quality adjustment. Serves as baseline.

- **Basic Hedonic:**

Standard hedonic regression. Independent Lasso model for each period. Controls for product features.

- **Basic + Error Feature:**

Hedonic regression with previous period prediction error as additional feature. Captures time dependencies.

- **Delta Model (Lasso):**

Directly models log price differences between consecutive periods using Lasso. Uses OOF predictions to avoid mean-matching artifacts. Automatic feature selection.

- **Delta + Error Feature:**

Combines Delta model with error feature from previous period pair. Captures both price changes and time dependencies.

- **OLS Delta Model:**

Directly models log price differences using OLS (no regularization). All features included. Simpler interpretation.

- **Time Dummy Model:**

Pools consecutive periods data together. Adds time dummy variable. Single model predicts both periods. More efficient parameter sharing.